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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

First Semester M.Tech. (EVT) (Credit System) Degree Examinations

December 2024 - January 2025

22EVT102 – VEHICLE DYNAMICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Elaborate roll of center of gravity and lumped load in vehicle dynamics.	08	L*2	1	1,3,5
	b) Elaborate various arbitrary forces acting on vehicle.	08	L3	1	1,3,5
	c) Explain Newton's second law.	04	L1	1	1,3,5
2.	a) Derive expression for dynamic load carried by front and rear wheel.	08	L2	1	1,3,5
	b) Derive expression for brake proportioning.	08	L1	1	1,3,5
	c) List the motion variable considered in vehicle dynamics.	04	L2	1	1,3,5
Unit – II					
3.	a) With neat diagram explain power train components with transverse weight shift.	08	L2	2	1,3,5
	b) Derive expression for traction limited tractive force	08	L4	2	1,3,5
	c) Write a note on auto gear mechanism.	04	L2	2	1,3,5
4.	a) Derive expression for stop time of brake.	08	L3	2	1,3,5
	b) Define brake factor and derive the expression for the same.	08	L1	2	1,3,5
	c) With neat diagram explain working of drum type brake.	04	L2	3	1,3,5
Unit – III					
5.	a) Explain various drag components and their effect on vehicle movement	08	L2	3	3,4
	b) Explain mechanics of air flow around the vehicle without air friction.	08	L2	3	3,4
	c) Elaborate the uses of bumper spoiler.	04	L1	3	3,4
			L2	3	3,4
6.	a) Explain the various types of vibrations occurring in tire.	08	L2	3	3,4
	b) Define slip and derive slip equation and comment on brake forces.	08	L1	3	3,4
	c) Explain hysteresis and adhesion property of tire dynamics.	04	L2	4	3,4
Unit – IV					
7.	a) With neat diagram explain active control in ride dynamics.	08	L2	4	1,3
	b) Elaborate effect of suspension nonlinearity in vehicle ride.	08	L3	4	1,3
	c) List the factors involved in preparation for ride.	04	L1	4	1,3
8.	a) With neat diagram elaborate fundamentals of ride dynamics system.	08	L2	4	1,3
	b) Explain road roughness effect on ride.	08	L1	4	1,3
	c) Elaborate basics of vehicle response properties.	04	L2	4	1,3
Unit – V					
9.	a) With appropriate example, elaborate active suspension methods.	10	L2	5	1,3
	b) Explain front wheel steering system forces and moments.	10	L2	5	1,3
10.	a) Explain various types of solid axel suspension.	08	L2	5	1,3
	b) Derive expression for force acting on rear wheel during acceleration.	08	L3	5	1,3
	c) Explain three link rear suspension.	04	L1	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
